

Integrated Reporting Quality: An Empirical Analysis

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70

ABSTRACT

Integrated Reporting (IR) represents the most recent evolution of the corporate reporting movement. Despite the proposed benefits to stakeholders and the number of contributes aimed at identifying best practices in its adoption, IR is still scarcely diffused among companies and when it is adopted the framework is not fully implemented. Starting from these premises, this paper aims to assess the quality of integrated reports issued by firms. We have developed a scoring model and an IR Scoreboard (IRS), which were applied to analyze 116 integrated reports issued in years 2013 and 2014. The main findings show that IR quality is low. Generally, firms follow the IR framework, but scarce information is disclosed on aspects such as capital, the business model, strategic priorities, and the value creation process; more attention is given to the IR form than to its content. Copyright © 2018 John Wiley & Sons, Ltd and ERP Environment

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Introduction

INTEGRATED REPORTING (IR) REPRESENTS A NATURAL EVOLUTION OF THE CORPORATE REPORTING MOVEMENT, AIMED AT FAVOURING THE implementation of a sustainable strategy by integrated thinking. It should present an organization's business model and value creation process, highlight the use of and dependence on different types of resources/capital, enable stakeholders to evaluate more effectively a firm's ability to create present and future value, and allow users of performance information to assess corporate long-term viability and more effectively allocate scarce resources. IR combines financial, governance, social, and environmental information in a single report.

Since the launch of the IR proposal, the literature has focused mainly on IR's role, objectives, and determinants, the main critical issues in its framework and implementation, and the relation with other types of disclosure. Despite the proposed benefits to stakeholders, as well as the number of contributes aiming at identifying best practices in its adoption, recently some studies have showed that IR is still scarcely diffused among companies and when it is adopted the framework is not fully implemented (de Villiers *et al.*, 2014; Global Reporting Initiative (GRI), 2013a). A recent study (International Integrated Reporting Council (IIRC), 2014) showed great interest in IR, even though the majority of the interviewed companies declared a reluctance and difficulties in fully implementing the IR framework. These mainly concerned the request of too much confidential information, technical issues in

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providing data, and the large amount of resources needed. In particular, issues on the complete application of the IR framework and the quality of integrated reports have been raised. This evidence has recently shifted the interest of practitioners (Ernst & Young, 2014; GRI, 2013b; KPMG, 2012; PricewaterhouseCoopers, 2014) and academics (Eccles and Krzus, 2014; Eccles and Saltzman, 2011; Eccles and Serafeim, 2011, 2013, 2015) from the type and quantity of information included in integrated reports towards their quality.

Starting from these premises, this paper aims to assess the quality of integrated reports issued by firms. This is a quite underdeveloped topic in the literature specifically focused on IR, while some relevant contributions may be found in studies on the quality assessment of corporate social responsibility (CSR) and sustainability disclosure. Thus, to assess the quality of integrated reports, we have referred to the literature on the quality of voluntary disclosure. In particular, we have developed a scoring model, referring to the framework proposed by Hammond and Miles (2004), and identified an IR Scoreboard (IRS), articulated into four areas. On the basis of the IRS, we have analyzed 116 integrated reports of 58 companies, taken from the Getting Started section with IR examples of the IIRC website, issued in years 2013 and 2014.

The main findings show that the quality of analyzed integrated reports is low. Generally, firms follow the IR framework, but scarce information is disclosed on relevant aspects such as capital, the business model, strategic priorities, and the value creation process. A limited depth of analysis characterizes all sections of the integrated report. More attention is given to the form than to the content of IR.

The remainder of the paper is organized as follows. The first section presents the main literature on IR and on the quality assessment of voluntary disclosure. The second section presents the research framework and methodology. The third section highlights and discusses the empirical findings. Finally, the last section reports some final remarks.

Literature Review

According to the IIRC (2013a), an Integrated Report is a ‘concise communication about how an organization’s strategy, governance, performance and prospects, in the context of its external environment, lead to the creation of value over the short, medium and long term’. IR provides ‘the holistic and integrated representation of the firm’s status according to the three performance perspectives: financial, social and environmental’. IR is much more than bringing together financial and sustainability reporting, as it provides comprehensive and comprehensible information about the organization’s total performance, prospective as well as retrospective. IR defines also a systemic representation of the firm’s performance, showing, evaluating, and controlling all the potential risks – financial and non-financial – that could damage the firm’s activities (Bebbington and Larrinaga, 2014). The IR framework is based on a wide concept of integration (Pistoni and Songini, 2015):

- the company performance is interpreted as the result of the combined use of different kinds of capital provided by different stakeholders: financial, human, technical, intellectual, social, and environmental capital;
- the framework requires to make explicit the relation between the firm’s strategy and performance;
- the integration is also pursued through organizational processes and procedures;
- the framework promotes integrated thinking and a strong collaboration among functions in order to improve the firm’s value creation in the long run;
- the value created is not only a result of the firm’s internal activities but also of the quality of the relationships with different stakeholders.

Previous research on IR has focused on the analysis of IR’s role and objectives (Beattie and Smith, 2013; Beck *et al.*, 2017; Brown and Dillard, 2014; Dumay *et al.*, 2016; Lodhia, 2014; Stacchezzini *et al.*, 2016), the IR framework’s main issues (Abeysekera, 2013; Haller and van Staden, 2014; Higgins *et al.*, 2014), and on the relations between IR and CSR and sustainability disclosure (Adams *et al.*, 2016; Maas *et al.*, 2016; Martinez-Ferrero *et al.*, 2015; Sierra-García *et al.*, 2015).

Recently, the literature has highlighted the determinants of IR focusing on why companies are producing integrated reports, and what are the explanatory factors for their adoption (Frias-Aceituno *et al.*, 2013; Jensen and Berg,

2012; van Bommel, 2014; Zabihollah, 2016). Other studies have analyzed main issues in the implementation of IR, by case study analysis (Burke and Clark, 2016; Havlová, 2015; Mio, 2016). Some authors have identified open questions in the adoption of IR (de Villiers *et al.*, 2014; de Villiers and Sharma, 2017; Flower, 2015; Morros, 2016; Rezaee, 2016).

Conversely, the literature on IR quality assessment is quite limited (Marx and Mohammadali-Haji, 2014; Ruiz-Lozano and Tirado-Valencia, 2016). In 2012, Ernst & Young launched the Excellence in Integrated Report Awards, aimed at analyzing the state of the art of IR in South Africa and at promoting better quality of IR and best practices in its adoption. In 2014, the IIRC itself published a document on 'Assurance on IR: an exploration of issues', with the aim of generating interest on quality issues. The Financial Reporting Council (2014) delivered 'Guidance on the Strategic Report', which focused on the quality of corporate reporting, in terms of its capacity to present the strategic elements that determine firm performance and value creation. Eccles and Krzus (2014, p. 191) stated that: 'it is not solely the absolute number of companies practicing integrated reporting, but the quality of adoption that matters ... Although companies may achieve a truly integrated report by other means, the effectiveness with which they apply these frameworks and standards will determine how useful these reports are to investors'.

The main critical issues on the quality of IR highlighted by the literature are: the absence of connectivity among strategy, the business model, performance and future outlook, due to the poor narrative flow and the limited use of diagrams and maps; the presence of an informative gap in areas such as governance, stakeholder engagement, and the materiality process; the inadequate description of the business model; and internal auditing, completeness of information, and limited third-party verification (Eccles and Serafeim, 2015).

In addition, some studies outlined that even though companies declare the application of the IR framework, they then implement it only partially. All such evidence has led the interest of practitioners, managers, and academics to shift from the type and quantity of information included in integrated reports towards their quality. In particular, quality refers to the capacity of IR to present the strategic elements that describe firm performance and value creation. It is 'a dimension linked to multiple characteristics and information areas that should be analyzed together to correctly appreciate its range' (Garegnani *et al.*, 2015, p. 542). 'Quality reporting is not about religiously complying with a framework, but about approaching strategic reporting with the right mindset. It's about taking a longer-term, broader, more operational perspective that will challenge how companies think, operate, monitor and report performance in a connected way' (PricewaterhouseCoopers, 2014).

Hence, quality assessment is not related only to the quantity of information reported. Some authors suggest that corporate disclosure should be informative and significant, understandable, comparable, comprehensive, material, reliable, and easily available (Garegnani *et al.*, 2015). According to Unerman (2000), to assess the quality of disclosure it is necessary to consider its contents as a whole. Other characteristics to be analyzed include disclosure's themes, range of issues and styles, nature, scope, coverage, time period, and location (Asif *et al.*, 2013; Hackston and Milne, 1996).

A number of guidelines have been proposed to assess the quality of CSR and sustainability voluntary disclosure. According to Hammond and Miles (2004), three main forms of assessment approaches exist: (1) formal monitoring procedures (professional scoring protocols, ranking exercises); (2) award schemes, aimed to highlight and reward best practices; and (3) academic quality assessment, which checks quality in different areas of disclosure, mainly using content analysis.

However, previous literature did not reach a consensus on which items to evaluate when assessing the quality of voluntary disclosure. The main reason can be traced in the difficulty to clearly and univocally define the concept of quality of disclosure. 'When a study covers voluntary disclosure, there is no predefined list of items, so it is necessary to define a list that will be applied to all of the companies surveyed' (Garegnani *et al.*, 2015, p. 546). So, usually the list of items is identified considering the different contributions of previous literature and regulatory frameworks (Adhikari and Tondkar, 1992).

With regard to the reasons explaining the quality of voluntary disclosure, three main theories can be taken into consideration. Namely, these theories give insight about the reasons underlying the low quality of such disclosure. First, the proprietary cost theory proposes that firms limit voluntary disclosure due to the existence of some disclosure costs including: data and information measurement, elaboration and communication, and the diffusion of strategic information which can give competitors an advantage (Dye, 1986; Verrecchia, 1983, 1990; Wagenhofer, 1990). Second, as highlighted by positive accounting theory and particularly in the light of the political

cost hypothesis, firms avoid disclosure of information in order not to attract the attention of politicians and regulators and to avoid political attacks (Watts and Zimmerman, 1986). Finally, according to **the agency theory**, there is a divergence between shareholder demand for information and managerial disclosure incentives. In fact, managers, especially risk-averse ones, are reluctant to reveal information because high disclosures can increase stakeholders' ability to discipline them (Nagar *et al.*, 2003).

Empirical Analysis

Research Objectives and the Quality Assessment Framework

In light of the literature on critical issues in IR adoption, and on the quality of voluntary disclosure (Eccles and Serafeim, 2015; Stubbs and Higgins, 2014), and considering the main theories concerning the firms' willingness to engage in voluntary disclosure, this paper aims to assess the quality of integrated reports, and positions itself in the 'academic quality assessment' stream of research (Hammond and Miles, 2004).

In particular, we posit two research questions (RQ 1 and RQ 2) and two hypotheses (HP 1 and HP 2):

RQ 1: What is the quality of integrated reports?

HP 1: *Integrated Reports issued by companies show a low quality*

RQ 2: Has the quality of IR improved during (over) time?

HP 2: *Over time IR quality improves but only incrementally*

In order to answer these questions and to test the hypotheses, we have firstly developed a framework for quality assessment of integrated reports. To define the list of items and build a scoring model, we have referred to the 11 attributes of the quality of CSR disclosure proposed by Hammond and Miles (2004): Quantitative disclosure, Third-party verification, Establishment of, and reporting against, (appropriate) targets, 'Warts and all' reporting, Adoption of reporting guidelines and standards, Ability to accurately assess performance from disclosure, Clear statement of vision from the CEO, Good coverage of significant issues, Wide access of the document, Reporting of normalized data, and Achievement of awards/accolades. Even though these attributes have been identified in the specific context of CSR disclosure, Hammond and Miles' framework includes not only the assessment of non-financial and qualitative information but also of financial and quantitative information, as the authors explain (p. 75): 'The CSR literature indicates that disclosure, which is quantitative and/or financial, 'warts and all' and comprehensive with respect to coverage would be classified as of a higher quality than discursive, self-laudatory, selected disclosure'. Moreover, the attributes of Hammond and Miles (2004) may be considered as broad guidelines suitable for the evaluation of the quality of both financial and non-financial (integrated) disclosure. Hence, we have linked the 11 attributes to four disclosure areas, articulated in 23 variables, which as a whole identify our IRS (Figure 1; Table 1):

- *Background area* evaluates whether the document presents an introduction devoted to discussing issues, such as: (1) motivations underlying the choice of adopting IR; (2) objectives pursued by IR; (3) beneficiaries of the document; (4) the manager in charge of the IR process; (5) the CEO's commitment; (6) the title of the report; and (7) consistency of IR with generally applied disclosure standards. Such aspects have been identified following the recommendations of the IR framework (IIRC, 2013b), as well as previous studies (Garegnani *et al.*, 2015).
- *Assurance and reliability area* assesses whether (1) an internal audit and/or (2) a third-party verification has been carried out; and (3) the company has received acknowledgements and awards for IR.
- *Content area* assesses the consistency of the document with the prescriptions of the IR framework regarding eight elements (organizational overview and external environment, business model, risks and opportunities, strategy and resource allocation, governance, performance, outlook, and basis of presentation) and two fundamental concepts (Capital and Value creation process).

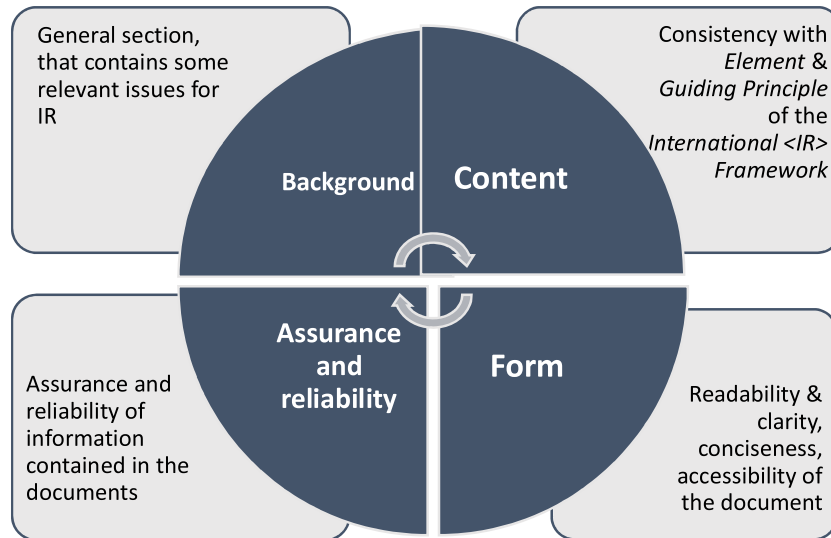


Figure 1. The integrated reporting scoreboard. [Colour figure can be viewed at wileyonlinelibrary.com]

Areas of the Integrated Reporting Scoreboard	Variables of the Integrated Reporting Scoreboard	Attributes of quality assessment (Hammond and Miles, 2004)
Background	1. motivations underlying the choice of adopting IR 2. objectives pursued by the IR 3. beneficiaries of the document 4. manager in charge of IR process 5. CEO's commitment 6. title of the report 7. consistency of IR with generally applied disclosure standards	1. Adoption of reporting guidelines and standards 2. Clear statement of vision from the CEO
Assurance and reliability	1. internal audit 2. third-party verification 3. acknowledgements and awards for IR	3. Third party verification 4. Achievement of awards/ accolades
Content	1. organizational overview & external environment 2. business model 3. risks & opportunities 4. strategy & resource allocation 5. governance 6. performance 7. outlook 8. basis of presentation 9. capitals 10. value creation process	5. Quantitative disclosure 6. Establishment of, and reporting against (appropriate) targets 7. 'Warts and all' reporting 8. Ability to accurately assess performance from disclosure 9. Good coverage of significant issues 10. Reporting of normalized data
Form	1. readability and clarity 2. conciseness 3. accessibility	11. Wide access of the document

Table 1. The relation between the areas and variables of the integrated reporting scoreboard and the attributes of quality assessment proposed by Hammond and Miles (2004)

- *Form area* assesses (1) readability and clarity of the document (presence of an index, graphs, tables, glossary, references to various sections of the document, and hyperlink to external sources, firm website or other documents); (2) conciseness (number of pages of the document); and (3) accessibility of the document (hard-copy documents versus website accessibility).

The 23 variables have been grouped in the four areas, trying to achieve the highest possible level of consistency with the literature on disclosure quality assessment (Botosan, 1997; Garegnani *et al.*, 2015), and cross-referencing the variables to the IR Guiding Principles of: Stakeholder Relationships (Background area); Reliability and Completeness (Assurance and Reliability area); Strategic Focus and Future Orientation, Connectivity of Information, and Materiality (Content area); and Conciseness, and Consistency and Comparability (Form area).

At the end of the process, we point out that some of the elements in our framework overlap with items previously proposed by the literature on the quality of financial disclosure (Botosan, 1997, p. 331) and codes of ethics (Garegnani *et al.*, 2015). For example, organizational overview and external environment, business model, strategy and resource allocation, performance, outlook, and value creation partly overlap with the major elements of Botosan’s DSCORE¹: Background Information, Ten or Five Year Summary of Historical Results, Key Non-Financial Statistics, Projected Information, Management Discussion and Analysis. On a different note, motivations underlying the choice of adopting IR, beneficiaries of the document, CEO’s commitment, assurance and reliability items, and form items (readability and clarity, conciseness, and accessibility) partly overlap with commitment from the top, style, and availability, relations with stakeholders, and compliance procedures in Garegnani *et al.*’s (2015) scoring model for Codes of Ethics’ quality assessment.

The following step was to develop a scoring system to assess each variable comprised in each of the four areas of the IRS. Quantitative scales have been defined to measure comparative positions and allow further analysis (Milne and Adler, 1999). Scoring systems have been extensively applied and validated in the fields of both CSR and sustainability, with reference to social and environmental reporting (Gray *et al.*, 1996; Romolini *et al.*, 2014), and in the field of accounting, concerning disclosure in financial reports (Botosan, 1997).

Concerning the Background area, the presence or absence of each of the seven proposed variables was evaluated. A score of 0 was given in the case of the absence of the item, while a score of 1 was given if the item was included in the IR document. The maximum score for this area is 7.

Referring to the Assurance and reliability area, a score of 0 was given in the case of the absence of each of the three items, while a score of 1 was assigned in the case of the presence of the item. The maximum score for this area is 3.

As for the Content area, to each of the ten variables was given a score between 0 (absence) and 5 (very high quality), according to the classification scheme presented in Table 2. Such a classification considers how the topic is presented, whether its description is exhaustive, and whether it refers explicitly to the IR guiding principles (strategic focus and future orientation, connectivity, stakeholder relationships, materiality, reliability and completeness, consistency and comparability). The maximum score achievable in the Content area is 50.

To each of the three elements of the Form area was given a score between 0 and 5, according to the classification schemes presented in Tables 3–5. The maximum score achievable in the Form area is 15. An overview of the scoring model is presented in Figure 2.

Score	Description
0	Content element absent
1	Content element present, but poor description and scarce reference to the IR guiding principles
2	Content element present; description based on some quantitative information and on a few IR guiding principles
3	Content element present; balanced description of contents; average quantity of information that refer to IR guiding principles
4	Content element present; good and detailed description of contents; many IR guiding principles considered
5	Content element present; excellent description of contents; quite all IR guiding principles used

Table 2. Scoring system of each variable of the content area

¹The purpose of my disclosure measure (which I refer to as DSCORE) is to produce a cross-sectional ranking of disclosure levels based on the amount of voluntary disclosure provided by firms in their annual reports.

Score	Readability and clarity of the document
0	Report quite not clear; absence of any element that can facilitate reading and comprehension of the document (i.e. graphs, tables, etc.)
1	Description mainly qualitative; scarce use of graphs and tables, that are not connected; absence of an index of the document, a glossary or a table with abbreviations
2	Adequate presence of graphs and tables; index with only few details
3	Graphs and tables facilitate the comprehension of the document; equilibrium between narrative flow and graphs/tables; references to other sections of the document avoid information redundancy
4	Very good use of graphs and tables; detailed index with hyperlinks; hyperlinks to external sources, to firm website or the other documents
5	Very good layout; index, graphs and tables clearly connected with the narrative qualitative flow

Table 3. Scoring system of the form area: Readability and clarity of the document

Score	Conciseness: n. of pages of the document
0	Not applicable
1	More than 200 pages
2	From 151 to 200 pages
3	From 101 to 150 pages
4	From 51 to 100 pages
5	Up to 50 pages

Table 4. Scoring system of the form area: Conciseness

Score	Accessibility of the document
0	Not applicable
1	Only hard copy document
2	Report available on the firm web site only as a pdf file
3	Report available on the firm web site, but not only as a pdf file: it is possible to access the contents directly from the firm web site
4	The report can be browsed on line, by surfing the firm web site, thanks to a highly interactive platform
5	Report contents highly accessible; the user can select issues and create a personalized report; highly interactive web platform; possibility to access the report by LinkedIn, Twitter, Facebook

Table 5. Scoring system of the form area: Accessibility of the document

Our IRS and scoring system have some similarities with the methodology used by Eccles and Krzus (2014) to assess the quality of reports issued by 124 listed companies. However, our study differs for the analyzed sample, which comprises both listed and not listed firms, and for the width of measurement scale (5-point scale, instead of 3).

The IIRC Database and the Analyzed Sample

The IIRC database includes integrated reports (sometimes annual reports), published since 2011 by companies from all over the world. We first examined all of the 128 integrated reports taken from the Getting Started section of the IR examples' website filtered with the years 2013 (63 reports available) and 2014 (65 reports) which were available as of December 2015. The Getting Started section, at the time of the study, included Integrated Reports considered suitable to illustrate IR guiding principles, content elements, and fundamental concepts and, therefore, they should be consistent with the IR Framework. To be able to make significant time comparisons, we left out the

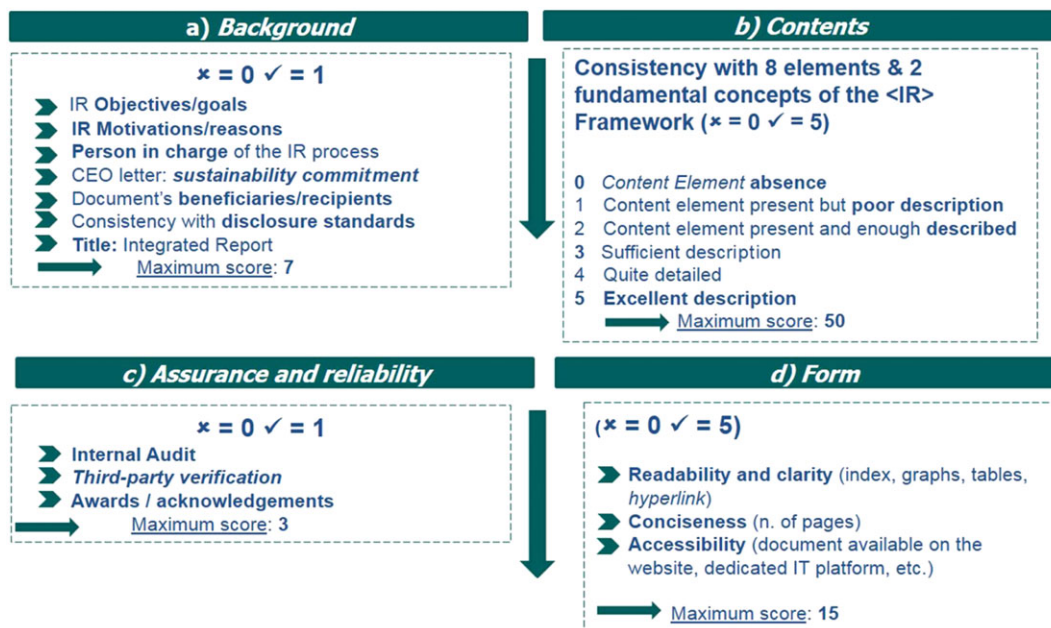


Figure 2. Overview of the scoring model. [Colour figure can be viewed at wileyonlinelibrary.com]

reports of firms whose reports were not available for both years. This resulted in the exclusion of 12 reports (7 firms have their report only in year 2014, 5 only in year 2013). Thus, our final sample consists of 116 reports (58 reports for each year analyzed). We have chosen the years 2013 and 2014 because our study was carried out during 2015. Moreover, in 2013, first, the 'Consultation Draft on the International Framework' (April; IIRC, 2013a) and, then, 'The International Framework' (December; IIRC, 2013b) were issued. Thus, we may suppose that reports published since 2013 are more consistent with the IR framework than in the past.

Table 6 presents the main features of the 58 analyzed companies. The 58 firms included in the sample operate in 12 different industries, only financial services being present with a percentage higher than 20%. With reference to the region of incorporation, Europe has the larger quota (54%).

Research Methodology

The data were collected and codified using visual content analysis (Weber, 1990) of all the 116 integrated reports of the 58 sample companies. Two researchers read carefully all 116 integrated reports and autonomously classified them consistently with the proposed scoring system. They were trained on the scoring protocol to assure the reliability of the analysis (Krippendorff, 1980). For the areas Background, Form, and Assurance and reliability, only few discrepancies emerged, which were discussed within the research group and resolved. During the process of analysis, the ten items of the Content area turned out to be the most controversial and seemed to involve more judgement by the two analysts. Hence, we decided to let the two analysts assign independently a score to each of the ten items and later controlled the inter-rater agreement (reliability of measurement) between the two analysts calculating the intra-class correlation coefficients (unreported, available upon request). The average intra-class correlation coefficient for the ten items is 0.82, ranging from a minimum of 0.74 to a maximum of 0.90, indicating a high degree of consistency between the two analysts. In light of the consistency of the independent scores, we considered in our analysis the average of the scores autonomously assigned by the two analysts.

We also tested if the different items in each area were actually measuring the same object and consequently if some redundant items should be removed. We used the KR20 test for categorical items (Background and Assurance and reliability areas) and Cronbach's alpha test for scale (0–5) variables (items in the Content and Form areas) (Table 7). The alpha level is never equal to or above 0.7 for all the four areas. This provides evidence that each of the 23 items contributes incrementally to the assessment of the quality of IR and therefore should be kept in the scoreboard. An alternative interpretation of this result, though, could be that we did not properly group the items

Organization type		
Publicly listed company	49	84%
Public Sector	5	9%
Other	4	7%
Total	58	100%
Region		
Europe	31	54%
Asia	10	17%
Africa	7	12%
South America	4	7%
Australasia	3	5%
North America	3	5%
Total	58	100%
Industry		
Financial Services	13	23%
Consumer Services	8	14%
Consumer Goods	6	10%
Basic Materials	6	10%
Industrials	6	10%
Oil and Gas	5	9%
Utilities	4	7%
Professional Services	3	5%
Healthcare	3	5%
Telecommunications	2	3%
Public Sector	1	2%
Real Estate	1	2%
Total	58	100%

Table 6. Main features of the sample (58 firms, 116 firm-year observations for year 2013 and 2014, balanced panel)

together in the four areas. We stick to the first interpretation (and reject the latter) because the variables measure different aspects and have been identified and grouped consistently with previous studies on disclosure quality assessment and with the IR Guiding Principles and Content elements (for the Content area). Moreover, for the assurance and reliability area (where we have the lowest alpha level equal to 0.09), it is reasonable to expect that single items would not be correlated: if an entity has its report externally audited, there is no reason to expect that it would be also audited internally or that it should receive an award for the quality of the report. On a different note, for the Background and Content areas, we may expect that an entity could disclose more diligently with regard to certain aspects, and less diligently on others (it may be also a deliberate tactic to disclose more on items which are less sensitive to have an excuse to disclose less on more sensitive ones, for the sake of conciseness). Finally, for the form area, where there is a (not negligible) alpha level of 0.42, the different elements of readability and clarity, conciseness, and accessibility not necessarily go together (a report may be very clear but too long, or may be very concise but obscure), even if they all refer to the form of the report and not the substance.

	Number of items	Type & Range	Alpha*
Background	7	Categorical (0-1)	0.61
Assurance & reliability	3	Categorical (0-1)	0.09
Content	10	Scale (0-5)	0.58
Form	3	Scale (0-5)	0.42

Table 7. Potential redundancy of items for each area

*Cronbach for scales, KR20 for categorical

We applied a methodology for the analysis of the scores consisting of the following steps:

- 1 describing the whole sample of 116 reports to get a general view of the advancement – at the time of the study – of the quality of IR with reference to the four areas that we identified;
- 2 comparing the relative total scores for each area (total average score divided by the maximum score achievable in the area), in order to evaluate which areas were best taken care of and which areas were relatively overlooked by the preparers of the reports;
- 3 observing the trend in each area, also considering the single items included in each of them, comparing the average scores for year 2013 and year 2014, and testing the statistical significance of the differences of the means (with a related samples (McNemar, 1947) change test for categorical variables, and with a paired samples t-test for scale variables).

Research Findings

To assess: (1) the quality of integrated reports in the four different areas: Background, Assurance and reliability, Content, and Form (in order to answer RQ 1 *What is the quality of integrated reports?* and test HP 1 *Integrated Reports issued by companies show a low quality*); and (2) changes in the quality of IR from 2013 to 2014 (in order to answer RQ 2 *Has the quality of IR improved during (over) time?* and test HP 2: *Over time IR quality improves but only incrementally*), we report, firstly, findings referring to the overall sample of 116 reports issued by the 58 companies and, then, we examine the differences between 2013 and 2014.

Analysis of the Whole Sample

Tables 8–10 describe the whole sample of 116 integrated reports. The elements in the Background section provide evidence of a situation that needs to be improved (Table 8). In particular, reporters paid little attention to the indication of the recipients (only 24%) and the reasons behind the adoption of IR (only 32%).

As for Assurance and reliability area, we observed that a large majority of the reports have been audited, internally (57%) and externally (71%), and that almost half of them (46%) received awards or acknowledgments for quality (Table 8).

Also the Content area shows considerable room for improvement: five items out of ten achieve an average score higher than or equal to 3 (i.e. they obtain a sufficient score). In detail, overview, risks, strategy, governance, and performance are on average fairly described in the reports, whilst that is not true for the business model, outlook, the basis of presentation, capital, and the value creation process (Table 9).

	NO	%	YES	%	Tot	%
Background						
Reasons	79	68%	37	32%	116	100%
Goals	67	58%	49	42%	116	100%
Recipients	88	76%	28	24%	116	100%
Responsible	61	53%	55	47%	116	100%
Commitment	50	43%	66	57%	116	100%
IR in title	73	63%	43	37%	116	100%
Framework standards	36	31%	80	69%	116	100%
Assurance & reliability						
Internal audit	50	43%	66	57%	116	100%
External audit	34	29%	82	71%	116	100%
Awards/acknowledgements	63	54%	53	46%	116	100%

Table 8. Presence of the items of the background and assurance and reliability areas in the overall sample (58 firms, 116 firm-year observations for year 2013 and 2014, balanced panel)

	Frequencies											Descriptives			
	0	0.5	1	1.5	2	2.5	3	3.5	4	4.5	5	Tot	Mean	Median	Stand. Dev.
Content															
Overview/Environment	0%	0%	0%	5%	9%	12%	23%	12%	17%	13%	8%	100%	3.35	3.25	0.97
Business model	1%	8%	4%	19%	8%	13%	14%	13%	12%	6%	3%	100%	2.59	2.50	1.23
Risk/opportunities	0%	3%	3%	8%	5%	10%	14%	23%	18%	12%	3%	100%	3.19	3.50	1.09
Strategy/Resource allocation	0%	0%	0%	13%	8%	16%	20%	17%	11%	12%	3%	100%	3.10	3.00	0.99
Governance	1%	1%	3%	9%	9%	16%	15%	12%	10%	22%	3%	100%	3.13	3.00	1.14
Performance	0%	0%	3%	8%	7%	15%	7%	23%	10%	22%	5%	100%	3.35	3.50	1.07
Outlook	1%	6%	10%	28%	9%	11%	9%	10%	4%	10%	1%	100%	2.32	2.00	1.22
Basis of presentation	19%	26%	6%	11%	3%	6%	4%	8%	3%	12%	1%	100%	1.70	1.00	1.60
The capitals	13%	6%	12%	21%	11%	9%	8%	7%	3%	8%	3%	100%	2.00	1.50	1.41
Value creation process	2%	6%	8%	10%	14%	16%	14%	17%	6%	4%	3%	100%	2.53	2.50	1.16
Form															
Readability and Clarity	0%		2%		12%		29%		44%		13%	100%	3.54	4.00	0.93
Conciseness	0%		25%		18%		32%		16%		9%	100%	2.66	3.00	1.27
Accessibility	0%		0%		7%		31%		47%		16%	100%	3.71	4.00	0.81

Table 9. Frequencies and descriptive statistics in the overall sample for the items in the content and form areas (58 firms, 116 firm-year observations for year 2013 and 2014, balanced panel)

	Range	Min	Max	Mean	Median	Stand. Dev.
Total Background Score	0-7	0	7.00	3.1	3	1.8
Total Assurance & Reliability Score	0-3	0.0	3.0	1.7	2.0	0.9
Total Content Score	0-50	12.0	39.5	27.3	27	5.5
Total Form Score	0-15	5.0	15.0	9.9	10.0	2.1

Table 10. Descriptives for total scores (58 firms, 116 firm-year observations for year 2013 and 2014, balanced panel)

These results suggest that, in general, the content of integrated reports still needs to be developed: when examining the specific issues covered, the richest information concerns items commonly disclosed in firms' financial reports, while the innovative topics of IR are still in their infancy.

In the Form area, some items perform fairly well (both readability/clarity and accessibility obtain an average score higher than 3.5 out of a maximum of 5), while conciseness – one of the novelty of IR – reaches on average only 2.66. Almost half of the companies (43%) achieve a score of 2 or less (Table 9).

At a glance, if we observe the total scores for the four areas, in Table 10, we see that the best-attended area is Form. The average total score for Form is 9.9 out of a maximum score of 15 (66%), followed by Assurance and reliability with an average of 1.7 out of 3 (57%), Content with an average score of 27.3 out of 50 (55%), and finally by Background which registers an average score of 3.1 out of 7 (44%).

In light of these research findings, we may say that our first hypothesis is verified, that is, the disclosure quality is still not satisfactory, especially in some areas. The pre-eminence of Form over substantial aspects such as Content may be linked – inter alia – to the high heterogeneity of results observed between the single items in the Content category, and in particular, as suggested above, to the fact that the most innovative parts of the IR framework still lack a truly effective disclosure.

We have not computed a total score summing up the total scores of the single areas, because the independence and heterogeneity of the four areas, and the difference in the measurement scales would make interpretation of the results unclear. For example, assurance and reliability, a very important aspect for evaluation of the quality of the reports, having a total maximum score of 3, would be underrepresented in a total score ranging from 0 to 75 made up from the sum of the total scores of the four areas.

Trend Analysis (2013 to 2014)

In this section, we examine the trend of disclosure in reports (from 2013 to 2014) separately for each area.

Background

Table 11 shows the percentage of occurrence of the seven items comprising the Background area in years 2013 and 2014, and the differences between the two years, which in general are not of relevant magnitude and, in any case, never statistically significant (as per a related samples (McNemar, 1947) change test).

The reasons, the goals, the responsible, and the level of commitment of integrated reports are disclosed in 2014 less frequently than in 2013. In general, the numbers are pretty disappointing, possibly because these items have been already declared in the past and, thus, the preparers of the reports might have thought that they did not need to repeat them in successive releases.

The recipients, the linkage to the IR Framework (Framework Standards), and the mention of IR in the title of the report, by contrast, appear in 2014 more frequently than in 2013. With reference to the first point (recipients), we may expect that, moving along with a truly 'integrated' approach, companies tend to adopt a view explicitly devoted to the materiality issue and, in doing so, they declare and identify more precisely the recipients of integrated reports.

The more frequent mention of the Framework and of IR in the title of the report could be related to the time effect (the Framework was launched in late 2013). In general, for the total score in the Background Area (Table 11) (as results from a paired-samples t-test), we observe a statistically not significant decrease from 2013 (mean = 3.17) to 2014 (mean = 3.00), $t(57) = -0.77$, $p = 0.445$ (two tailed). The mean decrease is -0.17 and the eta-squared statistic (0.01) suggests a small effect size (Cohen, 1988).

Background	2014	%	2013	%	2014-2013	%	<i>p-value</i> ^o
Reasons	17	29%	20	34%	-3	-5%	0.66
Goals	20	34%	29	50%	-9	-16%	0.15
Recipients	15	26%	13	22%	2	3%	0.80
Responsible	27	47%	28	48%	-1	-2%	1.00
Commitment	32	55%	34	59%	-2	-3%	0.85
IR in title	22	38%	21	36%	1	2%	1.00
Framework standards	41	71%	39	67%	2	3%	0.77
^o <i>p-value of Related Samples McNemar Change Test, testing the null hypothesis that there is not a significant change</i> ^{***} , ^{**} , ^{***} denote significance at .05, .01, and .001 level							
Background	2014	2013	2014-2013	<i>p-value</i>	<i>n</i> ²	<i>t</i>	<i>df</i>
Total average score (0-7)	3.00	3.17	-0.17	0.445	0.01	-0.77	57

Table 11. Presence of the items (%), and total score, background area, comparison 2014-2013

The relative lack of attention in this area may be explained by the intrinsic nature of the area itself, which provides the context for the information and in some cases may be taken for granted by preparers and users, at least in part, after the first release of an integrated report. Yet, the disclosure on specific themes is improving, such as the disclosure of recipients and the explicit mention of the Framework standards.

Assurance and Reliability

The results are similar in 2014 and 2013 (no change is statistically significant). In particular: (1) the majority of companies have their IR audited internally; (2) almost three out of four companies obtain an external auditor's opinion before formally releasing their reports; and (3) more than 44% of the companies received a specific award/acknowledgement for their Integrated Reports (Table 12).

In general, for the total score in the Assurance and reliability area (Table 12), we observe a statistically not significant (as results from a paired-samples t-test) increase from 2013 (mean = 1.72) to 2014 (mean = 1.74), $t(57) = 0.13$, $p = 0.9$ (two tailed). The mean increase is 0.02 and the eta-squared statistic (0.00) suggests a negligible effect size.

In absolute terms, this area seems well attended (average total score 1.74 out of a maximum score of 3 in 2014). This is consistent with the origin of the reports, which were extracted from the IIRC website's Getting Started section, which supposedly included, at the time of the study, examples useful to illustrate good practices in IR. It is reasonable to expect that these reports, more often than not, were audited and in some cases received acknowledgements for their quality.

Content

The scores for almost all the items of the Content Area (with just the exception of the Strategy/Resource Allocation) are higher in 2014 than in 2013, with considerable improvements for several items, statistically significant in six out of ten cases (Table 13). Most remarkable are the strong increases in the disclosure of the basis of presentation and the enrichment of the information regarding the value creation process.

Assurance & reliability	2014	%	2013	%	2014-2013	%	<i>p-value</i> ^o
Internal audit	32	55%	34	59%	-2	-3%	0.8
External audit	42	72%	40	69%	2	3%	0.8
Awards/acknowledgements	27	47%	26	45%	1	2%	1.0
^o <i>p-value of Related Samples McNemar Change Test, testing the null hypothesis that there is not a significant change</i> ^{***} , ^{**} , ^{***} denote significance at .05, .01, and .001 level							
Assurance & reliability	2014	2013	2014-2013	<i>p-value</i>	<i>n</i> ²	<i>t</i>	<i>df</i>
Total average score (0-3)	1.74	1.72	0.02	0.9	0.00	0.13	57

Table 12. Presence of the items (%), and total score, assurance and reliability area, comparison 2014-2013

At the same time, looking at the final situation in 2014, we notice that all the more innovative issues whose disclosure is requested by the IR Framework (e.g. business model, outlook, basis of presentation, capital, value creation process) still lag behind. In other words, even if we registered a good improvement in the level of disclosure of these issues, all of them achieve an average score lower than 3, and two of them (basis of presentation and capital) are distant by far from that threshold.

By contrast, items commonly reported in financial statements, such as performance, governance, and risks/opportunities, show average disclosure levels well above the threshold of 3. It appears that the truly innovative content of IR needs to be further developed by companies, even if we traced from 2013 to 2014 relevant improvements in the level of disclosure of some items.

At a glance, the total score reflects the positive changes registered in almost all of the items: we observe a statistically significant (as results from a paired-samples t-test) increase from 2013 (mean = 25.44) to 2014 (mean = 29.08), $t(57) = 5.155$, $p < 0.001$ (two tailed). The mean increase is 3.64 and the eta-squared statistic (0.32) suggests a large effect size (Table 13). Overall, the level of disclosure in 2014 is not yet satisfactory (the total score of 29.08 in 2014 corresponds to an average score for an item of the area of 2.9, still below the sufficiency threshold of 3).

The continuing process of collective learning, the need for the firms to adapt their organizational structure, their internal processes, and IT systems, in order to comply with the relatively new and complex IR Framework, may help explain the situation both in dynamic terms (the sensitive improvement between 2013 and 2014), and as it emerges in the last year analyzed (2014).

Form

All the items in 2014 score an average measure higher than in 2013, and all the changes are statistically significant (Table 14).

It is noteworthy that Conciseness (which is a key element of IR, due to its miscellaneous and extremely intertwined content), even if increased in 2014, reports an average measure of 2.8, still below the threshold for sufficiency of 3. On the other hand, clarity (which is another crucial issue, since integrated reports contain pieces of information which deal with highly complex and heterogeneous topics), as well accessibility are improving and they already achieve a score sensibly higher than 3.

As we expected after examining the score of the single items in the Form area, we observe a statistically significant (as results from a paired-samples t-test) increase of the overall score from 2013 (mean = 9.40) to 2014 (mean =

	2014	2013	2014-2013	
Content	<i>Average</i>	<i>Average</i>	<i>Average</i>	<i>p-value</i> ^o
Overview/Environment	3.4	3.3	0.1	0.55
Business model	2.8	2.4	0.4*	0.04
Risk/opportunities	3.4	3.0	0.4**	0.007
Strategy/Resource allocation	3.1	3.1	0.0	0.96
Governance	3.3	3.0	0.3*	0.03
Performance	3.4	3.3	0.2	0.35
Outlook	2.5	2.1	0.4	0.101
Basis of presentation	2.1	1.3	0.8**	0.001
The capitals	2.2	1.8	0.4*	0.02
Value creation process	2.9	2.2	0.7***	0.000

^o*p-value of paired sample t-test, testing the null hypothesis that there is not a significant change* *, **, *** denote significance at .05, .01, and .001 level

Content	2014	2013	2014-2013	<i>p-value</i>	<i>n</i> ²	<i>t</i>	<i>df</i>
Total average score (0-50)	29.08	25.44	3.64	0.000	0.32	5.155	57

Table 13. Scores on the single items and total score, content area, comparison 2014-2013

	2014		2013		2014-2013		
Form	Average		Average		Average		p-value ^o
Readability and Clarity	3.7		3.4		0.4 [*]		0.02
Conciseness	2.8		2.6		0.2 [*]		0.03
Accessibility	3.9		3.5		0.4 ^{**}		0.003
^o p-value of paired sample t-test, testing the null hypothesis that there is not a significant change ^{*, **, ***} denote significance at .05, .01, and .001 level							
Form	2014	2013	2014-2013	p-value	n ²	t	df
Total average score (0-15)	10.43	9.40	1.03	0.000	0.24	4.293	57

Table 14. Scores on the single items and total score, form area, comparison 2014-2013

10.43), $t(57) = 4.293$, $p < 0.001$ (two tailed). The mean increase is 1.03 and the eta-squared statistic (0.24) suggests a large effect size (Table 14).

The research findings concerning the trend of disclosure in reports (from 2013 to 2014) only partially support the second research hypothesis (that the disclosure quality is improving over time but only incrementally), as long as we observe statistically significant increases of the average scores combined with large effect sizes of the differences between 2014 and 2013 in the Content and Form areas, whilst there are no significant changes in the Background and Assurance and reliability areas.

Discussion

Overall, though, the quality of integrated reports is low. Reporters seemingly paid more attention to the form than to the content of the reports (the average total score in 2014 for Form is 3.5 compared with 2.9 for Content). This may be explained if, in the implementation of an IR project, the organizations have decided to concentrate their efforts on the easier task (devising the right form to present the information) and, only later on, have they redirected their efforts to the improvement of the content, which requires considerable resources to adapt internal processes, IT systems, etc.

Other main reasons that may explain the incomplete adoption of the IR framework and its low quality may be, first, the fact that the IR proposal, even though interesting and ambitious, today seems to be a bit idealistic and unfeasible. In fact, the relevant costs (both explicit and implicit) sustained by project implementation do not seem to be adequately covered by the associated benefits. Some authors proposed that perfectly credible (or completely unbiased) disclosure is not optimal because it is too costly (Core, 2001; Watts and Zimmerman, 1986). In addition, in light of today's general economic situation, the impact of overall costs represents a relevant issue for companies.

Second, theoretical studies have highlighted the risks of disclosure redundancy – above all regarding information potentially relevant for competitors – which may undermine both the firm's specificity and the distinctive factors that provide the foundations for the firm's competitive advantage (Graham *et al.*, 2005).

Moreover, it has to be taken into consideration the potential reaction of financial markets and stakeholders to the communication of some negative results, as well as their perception of low reliability of the disclosed information. Kasznik and Lev (1995) suggested that firms are penalized for disclosing bad news early.

With reference to the context, the implementation of the IR framework implies that agents operating inside the economic system recognize the stakeholder approach as an important component of firm strategy. If, on the one hand, this is a desirable situation, on the other hand, it may clash with today's dominant economic culture still very centred on the shareholder value model and on the company's financial results. Moreover, the stakeholders may not be prepared and proficient enough to understand the relevance of such a wide, systematic, and analytical disclosure.

Finally, another aspect preventing managers from voluntarily disclosing corporate information is driven by the so-called 'disclosure precedent': on the one hand, the firm has to maintain the same disclosure pattern along time,

although this may be difficult to preserve, on the other hand, disclosure precedent impacts on stakeholders' expectations concerning the consistency of the new disclosure along time, in both cases of good or bad news.

Concluding Remarks

The objective of this paper has been to assess the quality of integrated reports. Two research questions and two hypotheses have been proposed.

A framework for quality assessment of integrated reports has been developed, referring to Hammond and Miles (2014), which has been applied to evaluate: (1) how the IR Framework has been implemented by sample companies, and (2) the changes from 2013 to 2014.

Hypothesis 1 (HP 1: *Integrated Reports issued by companies show a low quality*) is supported by our research findings, while hypothesis 2 (HP 2: *Over time IR quality improves but only incrementally*) is only partially confirmed.

Our quantitative findings present evidence that even though a significant improvement of the quality from 2013 to 2014 in certain areas (namely Content and Form) can be observed, the overall quality of integrated reports is still quite limited particularly with regard to the Content and Background areas. In addition, firms seem to give more attention to the form than to the content of reports. It is noteworthy that the most innovative and distinctive elements of IR are those disclosed with less quality, while a higher quality characterizes aspects traditionally disclosed by firms, such as performance, governance, and risks. Similarly to Stubbs and Higgins' (2014) research, our findings seem to confirm that IR, being in its early adoption stage, may take more time before such an innovative disclosure mechanism emerges. They also suggest that currently IR represents a 'transition' (Laughlin, 1991) from sustainability reporting rather than a radical innovation driving transformation and that the lack of comprehensive standards may be inhibiting a more widespread adoption (Stubbs and Higgins, 2014).

Our findings are consistent also with the work of Ruiz-Lozano and Tirado-Valencia (2016, p. 252) that highlighted how 'despite the effort of the analyzed companies to address the guiding principles, they still have a long way to go especially in relation to the principle of conciseness'. In addition, we found similar results to Marx and Mohammadali-Haji (2014) who reported quality issues with regard to materiality, description of capital, and independent and full assurance of the reports. They outlined also that the description of risks was largely generic and that only few companies described their objectives and strategies.

To summarize, our empirical analysis highlighted the existence of some critical aspects related to the quality of integrated reports, showing that the whole application of the IR model is still far away. Difficulties in the complete application of the IR framework have been pointed out by the IIRC itself, which issued a document giving an account of comments received from institutional investors and companies. Communicating the firm strategy and the feature of the value creation process represents the most relevant difficulty for enterprises (IIRC, 2014), and this is also consistent with our results.

The results of this paper are aligned with different theoretical streams. First, they are consistent with proprietary cost theory, which suggests that managers prefer not to disclose information that may affect the firm's competitive position in the market, even if this would decrease the associated cost of capital (Dye, 1986). Second, as highlighted by Nagar *et al.* (2003) and Berger and Hann (2003), agency issues may explain reduced voluntary disclosure because managers desire to avoid potential attention from investors about issues that, if fully disclosed, can represent a threat for them. Third, according to Watts and Zimmerman (1986), political costs may also explain a reluctance in wider disclosure as managers prefer not to disclose voluntary information that regulators might use against the company and want to avoid being subject to political attacks (Alsaeed, 2006; Camfferman and Cooke, 2002).

To conclude, our study suggests that the methodological efforts and recommendations on IR should be directed towards the improvement of quality, especially with regard to reliability and selectivity of data and information provided, instead of promoting the expansion of the contents or more complex forms of disclosure. For these reasons, in the near future we can expect that the implementation of the IR approach will continue to be limited.

Moreover, a debate has to begin among companies, professionals, associations, and institutions on why the majority of firms implement IR with limited quality, and on what are the positive impacts, the real benefits versus the costs, but also the risks in terms of disclosing strategic and confidential information.

This paper contributes to the research on IR in different ways: first, it has proposed an original framework for the quality assessment of IR, the IRS; second, it has applied the IRS to assess the quality of a sample of integrated reports, thus giving a picture of the state of the art regarding the implementation of IR; and third, it has highlighted the main critical issues in the adoption of the IR framework, with particular regard to quality aspects.

However, our research has some limitations typical of qualitative studies, such as content analysis. Such a methodology may be characterized by an unavoidable subjectivity of the evaluation process (Hammond and Miles, 2004; Krippendorff, 1980) as well as the difficulty of generalizing the results beyond the sample studied. Even though the IRS and the scoring system have been developed by referring to previous studies, first their definition and then implementation to assess the 116 integrated reports may have implied some subjectivity. Another group of researchers could have developed a slightly different framework and scores or found slightly different findings using the same framework and scoring system.

Moreover, we analyzed only integrated reports available on the Getting Started section of the IR examples' website, which included, at the time of the study, reports considered suitable to be used as benchmarks by beginner IR preparers. For this reason, the sample may not be representative of the average firm issuing an IR. In addition, we have focused the analysis on a specific period of time. Considering other periods or a longitudinal analysis could be an issue for future research.

As IR is still evolving, our research findings suggest numerous potential areas for future research. With regard to methodology, future research could try to increase the number of companies of the analyzed sample and the time span of analysis, and to adopt both multivariate analysis and a case study methodology to go deeper in evaluating the quality of integrated reports.

Moreover, as a direct extension of this research, several issues could be further investigated. How IR quality varies across companies, industries, and time may represent relevant research questions to explore. What are the determinants of IR quality; both internal drivers – related to firms' specific characteristics – and external ones – coming from the environment where the companies operate – could be another interesting aspect to examine.

Finally, the evidence that the overall quality of IR is quite limited requires investigating if there is an optimal level of disclosure that has to be taken into consideration in the assessment of IR quality. Since this optimal level could be different for the firm and for stakeholders, other aspects that seem worthy of investigation are, on the one side, the tendency towards a convergence of the firm's and stakeholders' perspectives and, on the other side, from the firm's point of view, a deeper analysis of the trade-off between the costs and benefits of high disclosure. In particular, the trade-off between more costly administrative procedures to gather information and more risk from divulging a firm's idiosyncrasies, weighed against the potential benefits stemming from the lower cost of capital and steadier relationships with strategic partners and the community at large, could be another interesting research area.

References

- Abeysekera I. 2013. A template for integrated reporting. *Journal of Intellectual Capital* 14(2): 227–245.
- Adams CA, Potter B, Singh PJ, York J. 2016. Exploring the implications of integrated reporting for social investment (disclosures). *The British Accounting Review* 48(3): 283–296.
- Adhikari A, Tondkar RH. 1992. Environmental factors influencing accounting disclosure requirements of global stock exchanges. *Journal of International Financial Management and Accounting* 4(2): 75–105.
- Alsaeed K. 2006. The association between firm-specific characteristics and disclosure: the case of Saudi Arabia. *Managerial Auditing Journal* 21(5): 476–496.
- Asif M, Searcy C, dos Santos P, Kensah D. 2013. A Review of Dutch Corporate Sustainable Development Reports. *Corporate Social Responsibility and Environmental Management* 20(6): 321–339.
- Beattie V, Smith SJ. 2013. Value Creation and Business Models: Refocusing the Intellectual Capital Debate. *British Accounting Review* 45(4): 243–254.
- Bebbington J, Larrinaga C. 2014. Accounting and sustainable development: An exploration. *Accounting, Organization and Society* 39: 395–413.
- Beck C, Dumay J, Frost G. 2017. In Pursuit of a 'Single Source of Truth': from Threatened Legitimacy to Integrated Reporting. *Journal of Business Ethics* 141(1): 191–205.
- Berger P, Hann R. 2003. Segment disclosures, proprietary costs, and the market for corporate control. Working paper. University of Chicago and University of Southern California.
- van Bommel K. 2014. Towards a legitimate compromise? Accounting. *Auditing & Accountability Journal* 27(7): 1157–1189.

- Botosan CA. 1997. Disclosure level and the cost of equity capital. *The Accounting Review* 72(3): 323–350.
- Brown J, Dillard J. 2014. Integrated reporting: On the need for broadening out and opening up, Accounting. *Auditing and Accountability Journal* 27(7): 1120–1156.
- Burke JJ, Clark CE. 2016. The business case for integrated reporting: insights from leading, practitioners, regulators and academics. *Business Horizons* 59(3): 273–283.
- Camfferman K, Cooke TE. 2002. An analysis of disclosure in the annual reports of UK and Dutch companies. *Journal of International Accounting Research* 1: 3–30.
- Cohen JW. 1988. Statistical Power Analysis for the Behavioral Sciences, 2nd edition. Lawrence Erlbaum Associates: Hillsdale, NJ.
- Core JE. 2001. A review of the empirical disclosure literature: discussion. *Journal of Accounting and Economics* 31: 441–456.
- Dumay J, Bernardi C, Guthrie J, Demartini P. 2016. Integrated reporting: A structured literature review. *Accounting Forum* 40(3): 166–185.
- Dye RA. 1986. Proprietary and non-proprietary disclosures. *Journal of Business* 59(2): 331–366.
- Eccles RG, Krzus MP. 2014. The Integrated Reporting Movement: Meaning, Momentum, Motives, and Materiality. John Wiley & Sons: Hoboken, New Jersey.
- Eccles RG, Saltzman D. 2011. Achieving Sustainability Through Integrated Reporting, Stanford Social Innovation Review, Summer: 55–61.
- Eccles RG, Serafeim G. 2011. The Role of the Board in Accelerating the Adoption of Integrated Reporting. Conference Board. <https://www2.fhcxchange.conference-board.org/2Fpublications/2Fpublicationdetail.cfm%3Fpublicationid%3D2049%26centerId%3D1> [20 January 2017].
- Eccles RG, Serafeim G. 2013. A Tale of Two Stories: Sustainability and the Quarterly Earnings Call. *Journal of Applied Corporate Finance* 25(3): 8–19.
- Eccles RG, Serafeim G. 2015. Corporate and Integrated Reporting: A Functional Perspective (January 31, 2014). Chapter in Stewardship of the Future, edited by Ed Lawler, Sue Mohrman, and James O'Toole, Greenleaf. Available at SSRN: <https://ssrn.com/abstract=2388716> or <https://dx.doi.org/10.2139/ssrn.2388716>
- Ernst & Young. 2014. Integrated reporting. Tips for organizations on elevating value, EYGM Limited. [http://www.ey.com/Publication/vwLUAssets/EY-integrated-reporting-tips-for-organizations-on-elevating-value/\\$FILE/EY-integrated-reporting-tips-for-organizations-on-elevating-value.pdf](http://www.ey.com/Publication/vwLUAssets/EY-integrated-reporting-tips-for-organizations-on-elevating-value/$FILE/EY-integrated-reporting-tips-for-organizations-on-elevating-value.pdf) [20 January 2017].
- Financial Reporting Council. 2014. Guidance on the Strategic Report. <https://www.frc.org.uk/Our-Work/Publications/Accounting-and-Reporting-Policy/Guidance-on-the-Strategic-Report.pdf>
- Flower J. 2015. The international integrated reporting council: A story of failure. *Critical Perspectives on Accounting* 27: 1–17.
- Frías-Aceituno JV, Rodríguez-Ariza L, García-Sánchez IM. 2013. Is integrated reporting determined by a country's legal system? An exploratory study. *Journal of Cleaner Production* 44: 45–55.
- Garegnani GM, Merlotti EP, Russo AA. 2015. Scoring Firms' Codes of Ethics: An Explorative Study of Quality Drivers. *Journal of Business Ethics* 126: 541–557.
- Global Reporting Initiative. 2013a. 'The sustainability content of integrated reports – a survey of pioneers'. <https://www.globalreporting.org/resource/library/GRI-IR.pdf> [3 June 2014].
- Global Reporting Initiative. 2013b. Sustainability information for the 2013 Global Conference on Sustainability and Reporting. <https://www.globalreporting.org/resource/library/GRI2013-Sustainability-Information.pdf> [3 June 2014].
- Graham JR, Harvey CR, Rajgopal S. 2005. The economic implications of corporate financial reporting. *Journal of Accounting and Economics* 40: 3–37.
- Gray RH, Owen DL, Adams C. 1996. Accounting and Accountability. Prentice Hall Europe: Harlow.
- Hackston D, Milne MJ. 1996. Some Determinants of Social and Environmental Disclosures in New Zealand, Accounting. *Auditing & Accountability Journal* 9(1): 77–108.
- Haller A, van Staden C. 2014. The value added statement – an appropriate instrument for Integrated Reporting, Accounting. *Auditing & Accountability Journal* 27(7): 1190–1216.
- Hammond K, Miles S. 2004. Assessing quality assessment of corporate social reporting: UK perspectives. *Accounting Forum* 28(1): 61–79.
- Havlová K. 2015. What Integrated Reporting Changed: The Case Study of Early Adopters. *Procedia Economics and Finance* 34: 231–237.
- Higgins C, Stubbs W, Love T. 2014. Walking the talk(s): Organisational narratives of integrated reporting, Accounting. *Auditing & Accountability Journal* 27(7): 1090–1119.
- International Integrated Reporting Council. 2013a. Consultation draft of the international Framework. <http://integratedreporting.org/> [3 June 2014].
- International Integrated Reporting Council. 2013b. The International Framework. <http://integratedreporting.org/> [3 June 2014].
- International Integrated Reporting Council. 2014. Assurance on IR. Overview feedback and call to action. <http://integratedreporting.org/> [26 September 2014].
- Jensen JC, Berg N. 2012. Determinants of traditional sustainability reporting versus integrated reporting. *An institutionalist approach. Business Strategy and the Environment* 21: 299–316.
- Kasznik R, Lev B. 1995. To Warn or Not to Warn: Management Disclosures in the Face of an Earnings Surprise. *The Accounting Review* 70(1): 113–134.
- KPMG. 2012. Integrated Reporting Performance insight through Better Business Reporting, Issue 2. <https://assets.kpmg.com/content/dam/kpmg/pdf/2013/03/integrated-reporting-2012.pdf> [26 September 2014].
- Krippendorff K. 1980. Content analysis: an introduction to its methodology. SAGE Publications: Beverly Hills, CA.
- Laughlin RC. 1991. Environmental disturbances and organizational transitions and transformations: some alternative models. *Organization Studies* 12(2): 209–232.
- Lodhia S. 2014. Exploring the transition to integrated reporting through a practice lens: An Australian customer owned bank perspective. *Journal of Business Ethics* 129: 585–595.
- Maas K, Schaltegger S, Crutzen N. 2016. Advancing the integration of corporate sustainability measurement, management and reporting. *Journal of Cleaner Production* 133: 859–862.

- Martinez-Ferrero J, Garcia-Sanchez IM, Cuadrado-Ballesteros B. 2015. Effect of Financial Reporting Quality on Sustainability Information Disclosure. *Corporate Social Responsibility and Environmental Management* 22: 45–64, <https://doi.org/10.1002/csr.1330>
- Marx B, Mohammadali-Haji A. 2014. Emerging trends in reporting: An analysis of integrated reporting practices by South African top 40 listed companies. *Journal of Economic and Financial Sciences* 7(1): 231–250.
- McNemar Q. 1947. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika* 12(2): 153–157.
- Milne MJ, Adler RW. 1999. Exploring the reliability of social and environmental disclosures content analysis. *Accounting, Auditing & Accountability Journal* 12(2): 237–256.
- Mio C. 2016. Internal application of IR principles: Generali's Internal Integrated Reporting. *Journal of Cleaner Production* 139: 204–218.
- Morris J. 2016. The integrated reporting: A presentation of the current state of art and aspects of integrated reporting that need further development. *Intangible Capital* 12(1): 336–356.
- Nagar V, Nanda D, Wysocki P. 2003. Discretionary disclosure and stock-based incentives. *Journal of Accounting and Economics* 34(1/3): 283–309.
- Pistoni A, Songini L. 2015. New trends and directions in CSD: the integrated reporting. In *Sustainability Disclosure: State of the Art and New Directions*, Pistoni A, Songini L (eds). Emerald: Howard House, Wagon Lane, Bingley, UK, 81–105.
- PricewaterhouseCoopers. 2014. Corporate performance: what do investors want to know? http://www.pwc.com/et_EE/EE/publications/assets/pub/pwc-investor-survey-ir-september2014.pdf [16 February 2015].
- Rezaee Z. 2016. Business sustainability research: A theoretical and integrated perspective. *Journal of Accounting Literature* 36: 48–64.
- Romolini A, Fissi S, Gori E. 2014. Scoring CSR Reporting in Listed Companies – Evidence from Italian Best Practices. *Corporate Social Responsibility and Environmental Management* 21(2): 65–81. <https://doi.org/10.1002/csr.1299>
- Ruiz-Lozano M, Tirado-Valencia P. 2016. Do industrial companies respond to the guiding principles of the Integrated Reporting framework? A preliminary study on the first companies joined to the initiative. *Revista de Contabilidad – Spanish Accounting Review* 19(2): 252–260.
- Sierra-García L, Zorio-Grima A, García-Benau MA. 2015. Stakeholder Engagement, Corporate Social Responsibility and Integrated Reporting: An Exploratory Study. *Corporate Social Responsibility and Environmental Management* 22(5): 286–304. <https://doi.org/10.1002/csr.1345>
- Stacchezzini R, Melloni G, Lai A. 2016. Sustainability management and reporting: the role of integrated reporting for communicating corporate sustainability management. *Journal of Cleaner Production* 136: 102–110.
- Stubbs W, Higgins C. 2014. Integrated Reporting and internal mechanisms of change, Accounting. *Auditing and Accountability Journal* 27(7): 1068–1089.
- Unerman J. 2000. Methodological issues: Reflections on qualification in corporate social reporting content analysis. *Accounting, Auditing and Accountability Journal* 13(5): 667–680.
- Verrecchia RE. 1983. Discretionary disclosure. *Journal of Accounting and Economics* 5: 179–194.
- Verrecchia RE. 1990. Information quality and discretionary disclosure. *Journal of Accounting and Economics* 12(4): 365–380.
- de Villiers C., Sharma, U. 2017. A critical reflection on the future of financial, intellectual capital, sustainability and integrated reporting, *Critical Perspectives on Accounting*. In press.
- de Villiers C, Rinaldi L, Unerman J. 2014. Integrated Reporting: Insights, gaps and an agenda for future research, Accounting. *Auditing & Accountability Journal* 27(7): 1042–1067.
- Wagenhofer A. 1990. Voluntary disclosure with strategic opponent. *Journal of Accounting and Economics* 12(4): 341–363.
- Watts RL, Zimmerman JL. 1986. *Positive accounting theory*. Prentice-Hall: Englewood Cliffs, New York.
- Weber RP. 1990. *Basic Content Analysis*, 2nd edition. Sage: Thousand Oaks, CA.
- Zabihollah R. 2016. *Business Sustainability Performance, Compliance, Accountability and Integrated Reporting*. Greenleaf Publishing: Salts Mill, Saltaire, UK.

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